Marine reserves

A promising tool for fish management is the creation of marine protected areas (MPAs), which legally protect fish stocks and ecosystems. MPAs cover 8 percent of the world's oceans, but only around 3 percent of MPAs are "fully" or "highly protected" areas, such as "no-take zones," where fishing, extraction of materials, dumping, drilling, and dredging are banned. One meta-analysis of scientific studies showed that the volume of fish species is on average 670 percent greater in fully protected "no-take" marine reserves than in areas that have no protection, and 343 percent greater than in partially protected MPAs. No-take zones effectively preserve and restore damaged ecosystems, too; coral reefs in protected zones of the Pacific Line Islands recovered from an El Niño event within a decade, but those in unprotected areas did not. Some studies suggest that legally enforced reserves may even help replenish fisheries outside their borders.